THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241
Introduction to Mathematical Analysis

Tutorial 3

Question 1. Consider the sequence

$$x_n = \cos\left(\frac{n\pi}{2}\right) + \frac{1}{n}.$$

(a) Find the limits of each of the following subsequences:

$$(x_{4k}), \qquad (x_{4k+2}), \qquad (x_{2k+1}).$$

(b) Use your answer to determine whether (x_n) converges.

(c) Determine the set of all possible limits of convergent subsequences of (x_n). Justify why there are no others.

Question 2. Let (x_n) be a bounded sequence of real numbers, and let $L \in \mathbb{R}$. Suppose that *every convergent subsequence* of (x_n) converges to L.

Prove that

$$x_n \longrightarrow L.$$

Hint: Suppose instead that x_n does not converge to L. What does the negation of the definition of convergence tell you? Can you then apply the Bolzano–Weierstrass Theorem to a suitable subsequence?

Question 3. Let (x_n) be a sequence satisfying

$$|x_{n+1} - x_n| \leq \frac{1}{n^2} \qquad \text{for every } n \in \mathbb{N}.$$

(a) Prove that (x_n) is a Cauchy sequence.
Hint: For $m > n$, write

$$x_m - x_n = (x_m - x_{m-1}) + \cdots + (x_{n+1} - x_n).$$

(b) One might be tempted to believe that the weaker condition

$$|x_{n+1} - x_n| \longrightarrow 0$$

is enough to imply that (x_n) is Cauchy.
Show that this is false by considering

$$x_n = \sqrt{n}.$$

Question 4. For each of the following series, determine whether it converges or diverges. State clearly which test you use and justify all hypotheses of that test.

(a)
$$\sum_{n=1}^{\infty} \frac{n+4}{n^3+2n+1}.$$

(b)
$$\sum_{n=1}^{\infty} \frac{\sqrt{n^2+1}}{n+2}.$$

(c)
$$\sum_{n=1}^{\infty} \frac{1}{n^2+\sqrt{n}}.$$

For each part, try to identify the simplest useful comparison before doing detailed algebra.

Question 5. Consider
$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^{2/3}}.$$

(a) Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

(b) Let
$$a_n = \frac{(-1)^{n+1}}{n^{2/3}},$$
and form the grouped series
$$(a_1 + a_2) + (a_3 + a_4) + (a_5 + a_6) + \cdots.$$
Prove that the grouped series converges to the same value as $\sum_{n=1}^{\infty} a_n$.
Hint: If $s_n = \sum_{k=1}^{n} a_k$, what are the partial sums of the grouped series in terms of (s_n)?

(c) Why does part (b) *not* contradict the fact that rearrangements of conditionally convergent series can behave differently?
Explain precisely what is preserved when terms are grouped in consecutive blocks, and what is changed in a genuine rearrangement.

Question 6. Define a sequence (a_n) of positive numbers by
$$a_n = \begin{cases} 3 \cdot 2^{-n}, & n \text{ odd}, \\ 2^{-n}, & n \text{ even}. \end{cases}$$

Consider the series
$$\sum_{n=1}^{\infty} a_n.$$

(a) Compute
$$\frac{a_{n+1}}{a_n}$$
separately for odd and even n. Show that
$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

does not exist.

What does this mean for the Ratio Test in the form stated in the lecture?

(b) Compute

$$\lim_{n\to\infty} \sqrt[n]{a_n}$$

and use the Root Test to determine whether the series converges.

(c) What does this example illustrate about the statement "a convergence test is inconclusive"? Does inconclusive mean that the series diverges?